NOT MEASUREMENT SENSITIVE

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SUPERSEDING
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MILITARY SPECIFICATION

BUTTONS, INSIGNIA AND FINDINGS (U.S.M.A.), GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the requirements for buttons, insignia and findings used by the United States Military Academy.

1.2 Classification. Buttons, insignia and findings covered herein shall be as specified herein and the applicable military specification sheet (see 2.1, 3.1, 6.3 and supplement 1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

NN-P-71 - Pallet, Material Handling, Wood Stringer Construction, 2 Way and 4 Way
PPP-T-45 - Tape; Paper, Gummed, Reinforced and Plain (Sealing and Securing
PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-676 - Boxes, Setup
QQ-N-290 - Nickel Plating, Electrodeposited
QQ-S-365 - Silver Plating, Electrodeposited, General Requirements For
QQ-Z-363 - Zinc Base Alloy, Die Casting
TT-E-489 - Enamel, Alkyd, Gloss, (For Exterior and Interior Surfaces)
UU-P-553 - Paper, Wrapping, Tissue
V-T-276 - Thread, Cotton

MILITARY

MIL-F-495 - Finish, Chemical, Black for Copper Alloys
MIL-P-15011 - Pallet, Material Handling, Wood Post Construction, 4 Way Entry
(See supplement 1 for list of associated specification sheets)

STANDARDS

FEDERAL

FED-STD-151 - Metal, Test Methods
FED-STD-406 - Plastics: Methods of Testing
FED-STD-595 - Colors

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in preparing this document should be address to: Director, The Institute of Heraldry, US Army, Cameron Station, Alexandria, VA 22304-5050, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Document Order Desk, Building 4D, Section D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

THE INSTITUTE OF HERALDRY

B-13-12 - Attaching devices Heraldic, prong and clutch type

STANDARD

THE INSTITUTE OF HERALDRY

Standard color chip set

(Copies of specifications, standards, drawings and specification sheets required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

(Figure 1 is a miniature copy of an Institute of Heraldry drawing and is for information only.)

2.2 Non-Government publications. The following documents form a part of this document to the extend specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-B- 487 - Method for Measurement of Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section.

ASTM-D-2240 - Rubber Property-durometer Hardness

ASTM-D-3951 - Standard Practice for Commercial Packaging.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable MS sheet. In the event of any conflict between the requirements of this specification and the MS sheet, the latter shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection.

3.3 Standard sample. Unless otherwise specified (see 6.3), the contracting officer shall furnish a standard sample or approved guide sample which shall be used for matching color and finish only. Variations from this specification may appear in the standard sample or approved guide sample; however, in such cases the specification shall govern.

3.4 Material. The material shall be as specified herein. Recovered material shall be used to the maximum extent possible.

3.4.1 Solder.

3.4.1.1 Hard solder. Hard solder shall be a silver alloy solder having a melting point of not less than 1075 degrees F (580 c).

3.4.1.2 Soft solder. Soft solder shall be a tin-lead alloy having a melting point of not less than 375 degrees F (190 c)

3.4.2 Copper-base alloys. Copper base alloys shall be free from pits, scale (including red oxide), dents, cracks, nicks, deep scratches, segregations and foreign matter that will not be removed in later processing. The chemical composition of the applicable copper-base alloys shall be as shown in table I, when tested as specified in 4.6.1.

TABLE I. Chemical composition

Copper base alloys	Composition (percent)						
	Copper	Nickel	Lead (Max)	Iron (Max)	Manganese	Zinc	Others
Red brass	84-86		.05	.05		Rem	.15 Max
Low brass	78.5-81.5		.05	.05		"	.15 Max
Yellow brass	64-68.5		.15	.05		"	.15 Max
Free cutting brass	60-63		3.75	.35		"	.50 Max
Nickel silver	63-66.5	16.5-19.5	.10	.25	.50	"	
Gilding metal	94-96		.03	.05		"	.13 Max

3.4.3 Steel. Steel shall be low carbon, cold rolled and oiled.

3.4.4 Zinc. Zinc shall be 99.6 percent pure when tested as specified in 4.6.1.

3.4.5 Aluminum. Aluminum shall be 99.8 percent pure.

3.4.6 Gold for plating. Gold for plating shall be 24 karat.

3.4.7 Enamel.

3.4.7.1 Hard enamel. Hard enamel shall be a hard glass type vitreous enamel fused with metalic oxides to produce the desired shade and opacity or translucence.

3.4.7.2 Enamel (paint). Enamel (paint) shall conform to class A or B of Federal Specification TT-E-489 and the shade shall match white No. 17875 of FED-STD-595.

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3.4.7.3 Baking enamel. The enamel for subdued insignia shall be a flat synthetic, baking enamel pigmented to the specified shade of black, brown, or green. The shade shall match the following standards established in FED-STD-595 as follows:

Black - 37038
Brown - 20095
Green - 34096

3.4.8 Epoxy resin (stoning epoxy). Epoxy resin shall be a stoning epoxy and have a shore D hardness of 85 and incorporate an ultraviolet inhibitor.

3.4.9 Lacquer. Lacquer shall be a pale clear synthetic lacquer. The use of pigmented lacquers shall not be permitted.

3.4.10 Rivets. Rivets, when required, shall be made from any type brass and may be solid or hollow. The rivets may be an integral part of the device or may be soldered or fused in place. Rivets shall be securely peened or swaged to insure a tight and close fit between component parts.

3.5 Design. The embossed design of the buttons and insignia shall be an exact duplicate of the Government loaned hubs (see 3.9) from which the contractor's working dies shall be extracted. The contractor's working dies shall be tooled and polished to remove any dents, nicks, scratches, or other imperfections.

3.6 Construction.

3.6.1 Stamping, trimming and piercing. The stamping shall have a well defined die struck edge. The device shall be trimmed and when applicable pierced to the die struck outline. All edges shall be clean, smooth and free from burrs, drag, step and rough edges. The stamping, trimming and piercing operation shall not damage or distort the design or alter the shape of the item.

3.6.2 Soldering and electronic fusion. Unless otherwise specified on the applicable specification sheet, all soldering shall be accomplished using hard solder specified in 3.4.1.1. Joints shall be clean, smooth, strong and free from burned or reduced areas. There shall be no excess solder and all excess flux shall be removed. When hard solder is specified, hard solder specified in 3.4.1.1 shall be used. When tested as specified in 4.6.2.1 the soldered parts shall not separate. When tested as specified in 4.6.3.1 the prong shall not separate at the joint. Where soft solder is specified, soft solder specified in 3.4.1.2 shall be used. When tested as specified in 4.6.2.2 the soldered device or prong shall not separate from the button or insignia. When fused joints are used, the prong shall not separate at the joint when tested as specified in 4.6.2.3.

3.6.3 Attaching devices. Attaching devices shall be as specified on the applicable specification sheet.

3.6.3.1. Prong and clutch type. Prongs shall be made from nickel silver specified in 3.4.2. Unless otherwise specified on the applicable specification sheet, the prongs as shown on drawing B-13-12 (figure 1) shall be 5/16 inch plus or minus 1/64 inch (0.79 cm plus or minus 0.04 cm) in length and shall be driven, swaged, soldered or electronically fused in the locations specified in the applicable specification sheet. Clutches as shown on drawing B-13-12 (figure 1) shall be made from any type brass. When tested as specified in 4.6.3.1 it shall not be possible to remove the clutch from the prong without first depressing the release wings.

3.6.3.2. Prong (registering). Prongs shall be made from nickel silver specified in 3.4.2. The prongs shall be of the dimensions shown on the applicable specification sheet. They shall be positioned and attached as required by the applicable specification sheet.

3.6.3.3 Screw post type and thumb nut. Screw post and thumb nut attaching devices shall be made from free cutting brass specified in 3.4.2 and shall conform to the details and dimensions as shown on the applicable specification sheet. The screw post shall be hard soldered or electronically fused to the back of the insignia in the location shown on the applicable specification sheet. When tested as specified in 4.6.3.2 thumb nuts shall not bind on the post's threads.

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3.6.3.4 Screw post with tube screw. Screw post and tube screw attaching devices shall be made from brass specified in 3.4.2 and shall conform to the details and dimensions as shown on and the applicable specification sheet. The screw post shall be hard soldered or electronically fused to the back of the insignia in the location shown on the applicable specification sheet. When tested as specified in 4.6.3.3 the tube screws shall not bind on the post's threads.

3.6.3.5 Stem with backshell. Stem and backshell attaching devices shall be made from low or red brass specified in 3.4.2 and shall conform to the details and dimensions as shown on the applicable specification sheet. The stem shall be soldered or swedged to the backshell in the location shown on the applicable specification sheet using hard or soft solder specified in 3.4.1.1 or 3.4.1.2. The position of the stem shall be such that if a rod is inserted in the eye of the stem, the rod shall be parallel to the lines of the background of the design of the button. When tested as specified in 4.6.3.4, the soldered or swedged stem shall not break loss or become wobble.

3.6.3.6 Tunnel back with backshell. Tunnel back with backshell attaching devices shall be made from red brass specified in 3.4.2 and shall conform to the details and dimensions as shown on the applicable specification sheet. The position of the shank shall be such that if a rod is inserted in the eye of the shank, the rod shall be at right angles to the parallel lines on the design of the button. When tested as specified in 4.6.3.5 the button shall not abrade or break the sewing thread.

3.6.3.7 Tack back. Tack back attaching devices shall be made from low or red brass specified in 3.4.2 and shall conform to the details and dimensions as shown on the applicable specification sheet.

3.6.3.8 Toggles, twist pin or bodkin. Twist pin or bodkin toggle attaching devices shall be made from hard drawn nickel wire specified in 3.4.2 and shall conform to the details and dimensions as shown on the applicable specification sheet.

3.6.4 Auxiliary components. Auxiliary components shall be as specified on the applicable specification sheet.

3.6.4.1 Burrs. Burrs shall be made from nickel plated steel specified in 3.4.3 and shall conform to the details and dimensions as shown on the applicable specification sheet.

3.6.4.2 Chain (hat). Hat chains shall be made from anodized gold aluminum specified in 3.4.5 and shall conform to the details and dimensions as shown on the applicable specification sheet.

3.6.4.3 Chain (overcoat). Overcoat chains shall be made from nickel plated steel specified in 3.4.3 and shall conform to the details and dimensions as shown on the applicable specification sheet.

3.6.4.4 Buckle, breastplate and waistplate. breastplate and waistplate buckles shall be made from red brass specified in 3.4.2 and shall conform to the details and dimensions as shown on the applicable specification sheet.

3.6.4.5 Hook. When specified, the hook shall be made from brass specified in 3.4.2 and shall conform to the details and dimensions as shown on the applicable specification sheet.

3.7 Finish. The color and finish of the insignia, buttons and findings shall be as specified on the applicable specification sheet.

3.7.1 Plating. All plating shall be by electroplating methods. The plating shall be continuous and unbroken over the entire plated surface. There shall be no cut-through, shaded, peeled, or blistered plating.

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3.7.1.1 Gold plating. Gold plating shall be accomplished using gold specified in 3.4.6. Prior to plating, all surfaces shall be cleaned of all matter which might affect the gold plating. Unless otherwise specified, the use of white metal as an undercoating shall not be permitted. When tested as specified in 4.6.4.1 no visible chemical reactions, such as evolution of gases, shall appear. When tested as specified in 4.6.4.2 the average quantity of gold on buttons within a lot shall not be less than the amount specified in the applicable specification sheet. In addition, any one dozen buttons tested shall contain not less than 90 percent of the specified quantity.

3.7.1.2 Hot nickel underplating. When nickel plating is specified on the applicable specification sheet, the underplating shall not be less than 0.00025 inch. Testing shall be standard commercial. In case of a dispute, testing shall be in accordance with ASTM test method B487.

3.7.2 Enameling. Enameling shall be applied so as to be uniform in color, free from bubbles, foreign inclusions, cracking, crazing, or other defects which affect appearance. There shall be no overrunning of enamel.

3.7.2.1 Hard enameling. Hard enamel specified in 3.4.7.1 shall be charged, fired and stoned level with the dikes. The enamel shall then be lapped, felted or wheel polished to produce a glass like finish. No design elements shall be removed during the stoning process. There shall be no noticeable burn spots as a result of the firing.

3.7.2.2 Enamel (paint). Enamel specified in 3.4.7.2 shall be applied in accordance with the manufacturer's standard practices.

3.7.2.3 Baking enamel. When insignia is to be subdued, baking enamel specified in 3.4.7.3 shall be used. Prior to enameling, the entire insignia shall be given a black chemical finish conforming to MIL-P-495. When tested as specified in 4.6.5.1 and 4.6.5.2, the enamel shall not flake or chip and no color shall transfer to the cotton except for a slight trace of residue.

3.7.3 Epoxy resin (stoning epoxy). Epoxy resin (stoning epoxy) specified in 3.4.8 shall be applied so as to be uniform in color, free from bubbles, foreign inclusion or other defects which affect appearance. There shall be no overrun of epoxy. The epoxy shall be stoned level with the dikes and lapped, felted or wheel polished to produce a glass like finish. No design elements shall be removed during the stoning process. When tested as specified in 4.6.6 the epoxy shall have a shore hardness of no less than 85D.

3.7.4 Lacquering. Insignia items shall be lacquered as indicated on the applicable specification sheet. The dry lacquer film shall be continuous, level, adherent and free from lint, dust or other foreign matter. When tested as specified in 4.6.7.1 there shall be no change in the appearance of the clear lacquer surface except a slight overall yellowing of the highlights. When tested as specified in 4.6.7.2 the tissue paper shall not adhere to the clear lacquered surface.

3.7.5 Anodizing. Aluminum shall be anodized to a minimum depth of 0.0003 inch and dyed to the required color. Anodized items shall be sealed in a nickel acetate bath and when tested as specified in 4.6.8.1 no stain shall appear on the anodize.

3.8 Marking for identification. The contractor shall stamp his trade or other identifying mark legibly and inconspicuously on each item in the location indicated in the applicable specification sheet.

3.9 Government loaned property. Hubs will be loaned by the Government and shall be used to make the contractor's working dies necessary for one contract or order (see 3.5).

3.10 Workmanship. Buttons, insignia and findings shall be clean, well made, and shall meet the acceptable quality levels established by this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5 and 4.6.

4.4 First article inspection. Inspection and testing of the first article shall be made on a completely finished item for all provisions of this specification applicable to end product examination and tests.

4.5 Quality conformance inspection. Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated (see 6.4).

4.5.1 Inspection of components.

4.5.1.1 Testing of components. In addition to the quality assurance provisions of the subsidiary specifications and drawings, inspection shall be performed on components and materials listed in table II for the test characteristics shown therein. The inspection level shall be S-1 and the requirements shall be applicable to the lot average.

TABLE II. Component testing

Component and lot expressed in terms of	Characteristic	Rqm't para	Test method	Rqmt appl to lot avg	#determ per sample unit	Results reported as	Sample Unit
Copper base alloys, as applicable (1 pound)	Chemical composition	3.4.2 and table I	Method 111 of Fed Std No. 151	X	2 for each element (composite)	To nearest 0.1%	One 4x4 sheet or equivalent for each alloy

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4.5.1.2 Certification of compliance. Components and materials listed below may be accepted on the basis of a contractor's certification of compliance for requirements specified in the applicable paragraphs of this specification.

Component	Characteristic	Rqm't para
Hard solder	Material identification	3.6.2
Soft solder	Material identification	3.6.2
Prong and clutch	Material identification	3.6.3.1
Screw post with thumb nut	Material identification	3.6.3.3
Screw post with tube screw	Material identification	3.6.3.4
Tunnel back with backshell	Material identification	3.6.3.6
Gold for plating	Material identification and karat	3.7.1.1
Quantitative gold	Material identification and karat	3.7.1.1
Baking enamel	Material identification	3.7.2.3
Epoxy resin	Material identification	3.7.3
lacquer	Material identification	3.7.4

4.5.2 Inspection of the end item.

4.5.2.1 Visual inspection of insignia. Examination shall be made at a distance of approximately 16 to 22 inches. The defects found during examination shall be classified in accordance with 4.5.2.1.1 and 4.5.2.1.2. The lots shall be inspected in accordance with 4.5.2.1.3. The unit of product for the examinations in 4.5.2.1.1 and 4.5.2.1.2 shall be one completely fabricated button or insignia as applicable.

4.5.2.1.1 Examination of insignia for defects in finish, design, material, construction, workmanship and marking. Defects designated by an asterisk (*) shall be classified as major when seriously affecting serviceability or appearance and minor when affecting appearance or serviceability but not seriously.

Examine	Defect	Classification	
		Major	(*) Minor
Color and finish (as applicable)	Not as specified	x	
	Does not compare favorably with standard or approved sample		*
	Buff, drag, blushed or cloudy finish on metal items		*
	Not highlighted (burnished) to the same extent as the standard or approved sample		*
	Crack, nick, chip, bubble or other imperfection of jewel		*
	Foreign matter embedded		*
	Surface contains pits, scale, nicks, scratches cracks or machine marks		*
	Not clean		*
Plating	Not plated when required	x	
	Not type specified	x	
	Not smooth, continuous or adherent	x	
	Cut-through or porous	x	

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Examine	Defect	Classification		
		Major	(*)	Minor
Enamel	Not enameled when required	X		
	Wrong color or shade	X		
	Cracked, chipped, or crazed		*	
	Bubbles, lumps, or foreign inclusion		*	
	Not stoned smooth (when required), e.g., is course or uneven		*	
	Not level with dikes, e.g., has high or low spots		*	
	Holes or blisters		*	
	Laps, overruns or skips		*	
	Burnt		*	
	Fishscale, discoloration, scumming or lack of gloss		*	
Lacquer	Not lacquered when required	X		
	Forms noticeable runs or contains sharp coarse particles		*	
	Hazy, cloudy or powdering		*	
	Not smooth, uniform, continuous or adherent, e.g., flaking, blistering, peeling or has runs sags or drops		*	
	Not set to touch			X
Design	Detail altered, does not conform to Government hub, drawing or standard sample	X		
	Any significant detail not clear, missing or obliterated, i.e., does not compare favorably with standard sample	X		
	Warp, twist or distortion producing irregular surface, contour or outline	X		
	Detail struck over resulting in a double impression	X		
	Area not pierced as indicated by drawing or standard sample	X		
	Not dapped when required	X		
	Dapping is not within specified radius		*	
Type of Class	Not as specified	X		
Material	Not as specified	X		
Workmanship and Construction	Surface or edge not clean and smooth, e.g., burr, drag, step or roughness		*	
	Metal marks on exposed surface, such as nick, dent, dig, gouge or scratch	X		
	Pinhole, crack or rupture		*	
	Die-struck edge, rim or outline is not well defined	X		
	Not accurately trimmed to die-struck edge	X		
	Dikes are distorted	X		
	Pierced edge not clean and smooth, e.g., any burr, drag, cutter or file mark		*	
Assembly	Sharp edge, burr or fin appearing on front, edge or back of button		*	
	Rolled edge not flush with back			X
	Looseness, i.e., shell edge not closed in, clinching the back	X		
	Edge cracked or split	X		

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Examine	Defect	Classification		
		Major	(*)	Minor
Stem	Missing, out of position, or is not type specified	x		
	Eye of hopper back not formed as required		*	
	Bent, loose, deformed or otherwise impaired		*	
	Sharp edge, fin or burr in eye of hopper back		*	
	Eye of stem improperly positioned by more than 10 degrees		*	
Threaded Post	Missing	x		
	Bent, loose, deformed or otherwise impaired		*	
	Thread stripped, flattened or damaged	x		
	Not as specified			x
Toggles, Bodkin	Toggle or cap missing	x		
	Not type specified			x
	Cap loose		*	
	Deformed, bent, twisted or otherwise damaged		*	
	Sharp edge, fin burr		*	
Screw tube	Threads stripped, flattened or damaged	x		
	Not type specified			x
	Opening obstructed, will not pass threaded post of corresponding size	x		
	Bent, damaged or defective affecting serviceability		*	
Washer, button	Bent, damaged or defective affecting serviceability		*	
Toggles, twist pin	Deformed, bent, twisted or otherwise damaged		*	
	Not type specified			x
	Sharp edge, burr or sliver		*	
	Loose fit to shank, i.e., may become detached in use		*	
Soldering	Solder does not completely unite component part to base metal, i.e., any perceptible opening between component parts	x		
	Joint not clean and not smooth, i.g., flux or excess solder not removed		*	
	Area burned or reduced in soldering			x
	Solder spatter on back		*	
Attaching or fastening device, prong and clutch type	Will not function or requires abnormal force to operate	x		
	Not type specified	x		
	Not positioned as specified		*	
	Clutch fastener will not engage or is loose fit on prong		*	
Marking	Missing or not as specified			x
	Illegible or misspelled			x
	Not inconspicuously placed where required		*	

4.5.2.1.2 Examination of buttons, insignia, and findings for dimensions. Any dimension which is not within the specified tolerances shall be classified as a defect.

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4.5.2.1.3 Inspection levels and acceptable quality levels. The inspection levels and acceptable quality levels (AQL's) expressed in defects per hundred units shall be as follows:

	Inspection level	AQL Major	AQL Total
For examinations applicable to 4.5.2.1.1	II	2.5	15.0
For examinations applicable to 4.5.2.1.2	S-2	One class	4.0

4.5.3 End item testing. Testing of the completed fabricated buttons, insignia, and findings shall be performed in accordance with Table V for the characteristic shown therein. The inspection level shall be S-1 and the AQL shall be 6.5.

Table V Testing of the End Item

Characteristic	Reqmt Para	Test Method	No. Determ Per Sample Unit	Results Reported as
Solder	3.6.2	4.6.2	1	Pass or fail
Prong/Clutch	3.6.3.1	4.6.3.1	1	Pass or fail
Screw Post/ Thumb Nut	3.6.3.2	4.6.3.2	1	Pass or fail
Screw Post/ Tube Screw	3.6.3.3	4.6.3.3	1	Pass or fail
Stem w/Back Back Shell	3.6.3.5	4.6.3.4	1	Pass or fail
Tunnel Back w/Back Shell	3.6.3.6	4.6.3.5	1	Pass or fail
Gold Plating (acid test)	3.7.1.1	4.6.4.1	1	Pass or fail
Gold Plating (quantitive)	3.7.1.1	4.6.4.2	1 (per dozen buttons)	To nearest 0.1 grain
Baking Enamel	3.7.2.3	4.6.5	1	Pass or fail
Epoxy	3.7.3	4.6.6	1	Pass or fail
Lacquer	3.7.4	4.6.7	1	Pass or fail
Anodizing	3.7.5	4.6.8	1	Pass or fail

4.5.4 Examination for count of buttons, insignia and findings in intermediate containers. Buttons, insignia or findings packaged for shipment shall be examined to determine conformance with package markings and specified quantity. The sample unit for this examination shall be one box (interior package). Any box containing less than the specified or marked quantity of buttons, insignia or findings shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units. The lot size shall be the number of intermediate containers.

4.5.5 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking requirements are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery, with the exception that it need not be sealed. Defects of closure listed below shall be examined on a shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

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EXAMINE	DEFECT
Markings, exterior and interior	Missing, incorrect, illegible or improper size, location sequence or method of application.
Materials	Component missing, damaged or otherwise defective, affecting serviceability.
	Insignia or buttons not mounted on cards (when required).
Cards not enclosed in antitarnish tissue paper.	
	Intermediate boxes missing or not specified type.
	Box not sealed with gummed paper tape.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, inadequate stapling.
	Bulging or distortion of containers.
Weight	Net weight exceeds the requirements.

4.6 Tests.

4.6.1 Chemical composition tests. Chemical composition shall be determined in accordance with Test Method III of FED-STD 151. Results shall be evaluated to determine compliance with the requirements specified in 3.4.2 and 3.4.4.

4.6.2 Soldering.

4.6.2.1 Test for hard soldered joints. Hard soldered insignia shall be placed in an oven maintained at 1065 degrees F., plus or minus 5 degrees F., for 15 minutes. While at this temperature the insignia shall be lifted by the attaching device and shall be inspected to determine compliance with the requirements specified in 3.6.2.

4.6.2.2 Test for soft soldered joints. Soft soldered insignia shall be placed in an oven maintained at 365 degrees F., plus or minus 5 degrees F., for 15 minutes. While at this temperature, the insignia shall be lifted by the superimposed device and shall be inspected to determine compliance with the requirements specified in 3.6.2.

4.6.2.3 Test for fused joints. Item to be tested shall be anchored on a horizontal surface. All prongs shall be grasped at least 1/3 its length above the fused joint and bent through an angle of 90 degrees (45 degrees to each side of vertical) until the prong breaks. The fused joint shall then be examined to determine compliance with 3.6.2. A bending tool in the form of a 45 degree template may be used for this test provided the prong is gripped at least 1/3 its length above the fused joint.

4.6.3 Attaching devices.

4.6.3.1 Test for prong and clutch type. The clutches shall be removed and replaced 10 times from the prong using the clutch release wings. An attempt shall then be made to remove the clutch by hand without first pressing the release wings to determine compliance with the requirements specified in 3.6.3.1.

4.6.3.2 Test for screw post type and thumb nut. The thumb nut shall be assembled on the screw post, during which, the operation of screw threads shall be observed to determine compliance with the requirements specified in 3.6.3.3.

4.6.3.3 Test for screw post type and tube screw. The tube screw shall be assembled on the screw post, during which, the operation of screw threads shall be observed to determine compliance with the requirements specified in 3.6.3.4.

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4.6.3.4 Test for stem with backshell. The stem with backshell shall be placed in an appropriate testing device so the stem is maintained in a vertical position. A steadily increasing tension load shall be applied directly upon the upright stem until the eye is noticeably elongated. Remove the compression load and examine the joint for failure to determine conformance with 3.6.3.5.

4.6.3.5 Test for tunnel back with backshell. Tunnel back buttons shall be hand sewn to any cloth material with four double cotton threads conforming to type IA3 or IB3, 4 ply of V-T-276. The sewn button shall be twisted 180 degrees in opposite directions, five times in each direction and then examined to determine compliance with 3.6.3.6.

4.6.4 Gold Plating.

4.6.4.1 Acid test for gold plating. The test acid shall be applied as follows: Place a drop of acid of not less than 1/16 inch (0.16 cm) in diameter on three different spots on the obverse of the plated surface allowing the drops to remain for not less than one minute, during which time the surface of the insignia shall be inspected to determine compliance with the requirements of 3.7.1.1. The test acid shall be applied at room temperature and shall consist of a solution containing 50 percent by volume of CP nitric acid (specific gravity 1.42) and an equal volume of distilled water. If the test insignia has been lacquered. the lacquer shall be removed prior to testing.

4.6.4.2 Quantitive test for gold plating. Standard commercial laboratory test methods shall be used to determine conformance of 3.7.1.1. In case of content dispute, the procedure specified in 4.6.4.2.1 shall be used.

4.6.4.2.1 Quantitative gold content test. To separate the gold layer from the base metal, dissolve twelve buttons in diluted nitric acid (1 volume of concentrated nitric acid and 4 volumes of water). Use only as much of the diluted acid as is necessary to dissolve the base metals. To facilitate solution of the base metals, the article being tested may be cut into pieces. When solution of the base metal is complete, catch the gold pieces on a filter, wash them, and transfer them to a 200 ml. beaker. It may be possible on occasions to avoid filtration by washing the gold by decantation. In either event, dissolve the gold by treating it with 10 ml. of diluted aqua regia (1 volume of concentrated nitric acid, 4 volumes of concentrated hydrochloric acid, and 6 volumes of water). If the filter paper is present, dilute the solution some-what, filter to remove paper pulp, wash and filter the paper pulp thoroughly, and evaporate the resulting solution to dryness on the steam bath. Take up the residue in 1 ml. of diluted aqua regia and 1 ml. of concentrated hydrochloric acid, and dilute the solution to 50 ml. Heat the gold solution on the steam bath, add 25 ml. of a freshly prepared solution of sulphurous acid (water saturated with sulphur dioxide), and allow to stand overnight to complete the precipitation and coagulation of the gold. Catch the precipitated gold on a smaller filter, wash it with diluted hydrochloric acid and 99 volumes of water), ignite the filter with its contents, and weight the residue as metallic gold.

4.6.5 Baked enamel.

4.6.5.1 Knife test. Draw a knife edge across the face side of the specimen marking four parallel scores 1/16 inch (0.16 cm) apart. Repeat this process drawing the knife perpendicular to the previously scored lines, making small blocks in the finish. The finish shall then be examined to determine compliance with 3.7.2.3.

4.6.5.2 Acetone test. The enameled surface shall be wiped five times with a piece of cotton saturated with acetone. The cotton shall be examined to determine compliance with 3.7.2.3.

4.6.6 Test for Shore hardness of Epoxy. Shore hardness shall be determined in accordance with ASTM 2240. Results shall be evaluated to determine compliance with the requirements of 3.7.3.

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4.6.7 Lacquer.

4.6.7.1 Liver of sulfur test. The samples to be tested shall be immersed in a 2 percent by weight of CP liver of sulphur (potassium sulfide) water solution, at a temperature of 100 F. to 110 F. for a period of three minutes. The samples shall be removed and rinsed in warm, then cold, then hot water, after which it shall be wiped gently with an absorbent cellulose material or whirled to remove residual moisture, allowed to air dry at room temperature (60 to 80 degrees F.) for a period of one hour and them examined to determine compliance with 3.7.4.

4.6.7.2 Tackiness test. At room temperature (60 to 80 degree F) press a piece of tissue paper against the lacquered surface for 15 seconds, using any pressure capable of being exerted between thumb and two fingers, after which the pressure shall be released and the item inspected to determine compliance with 3.7.4.

4.6.8 Anodizing.

4.6.8.1 Dye stain test. A spot of phosphate-free authorguinone violet RN dye shall be placed on the anodized surface. After five minutes the dye shall be wiped off and the surface examined to determine compliance with 3.7.5.

5. PACKAGING

5.1 Preservation. Preservation shall be levels A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Buttons, insignia and findings shall be prepackaged by being card mounted and wrapped, individually wrapped, or bulk wrapped in the quantities as specified in the applicable specification sheet. The mounting cards shall be a white sulphite board having a minimum thickness of 0.011 inch (0.027 cm) and be the approximate size indicated in the applicable specification sheet. The wrapping shall be anti-tarnish white tissue paper conforming to type II, grade A, class 1 of Federal Specification UU-P-553.

5.1.1.2 Intermediate packaging. The prepackaged buttons, insignia and findings shall be packaged in quantities as specified on the applicable specification sheet in a snug-fitting folding paperboard box conforming to style III, type B, class A of Federal Specification PPP-B-566; or setup paperboard box conforming to type I, grade A, class A, style 4 of Federal Specification PPP-B-676. Box closure shall be secured with 2 inch minimum width gummed paper tape conforming to type III of Federal Specification PPP-T-45 applied at the center of the length opening and extending along the bottom and up the sides at least two inches

5.1.2 Commercial. Buttons, insignia and findings shall be preserved and packaged in accordance with the applicable requirements of ASTM-D-3951.

5.2 Packing. Packing shall be level B or Commercial, as specified (see 6.2).

5.2.1 Level B. Buttons, insignia and findings of one type and class only, packaged as specified in 5.1 shall be packed in a snugfitting fiberboard shipping container conforming to style RSC of Federal Specification PPP-B-636. Closure shall be in accordance with the appendix of the container specification. The weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.2 Commercial. Buttons, insignia and findings of one type and class only, packaged as specified in 5.1 shall be packed in accordance with the applicable requirements of ASTM-D-3951.

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5.3 Palletization. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be palletized on a 4-way entry pallet in accordance with load type 1A of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, Type I, Class 1, Style 1, Size A, Wood Group I, II, III, or IV of MIL-P-15011, or, 4-way, Type IV, V or VIII, Class 1, Style A, Size 2, Wood Group I, II, III or IV, Grade A of NN-P-71. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking. In addition to any special markings required by the contract or purchase order, unit packages, intermediate boxes, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The buttons, insignia and findings are used with cadet uniforms of the United States Military Academy.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Type and class required (see applicable specification sheet).

d. When preproduction sample is required (see 3.2).

e. Whether standard sample is to be furnished (see 3.3).

f. Color of Type I, Class 2 Insignia (see applicable specification sheet).

g. Inspection record details (see 4.1).

h. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Samples. For access to guide samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first production sample or a standard production item from the contractor's current inventory as specified in 4.4. The first article should consist of one completely fabricated button. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.9

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6.6 Subject term (key word) listing.

Branch
Breast Plate
Buckle
Burr
Cadet
Cap
Chain
Cuff
Eyelet
Hat
Line
Toggle
Waist Plate

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

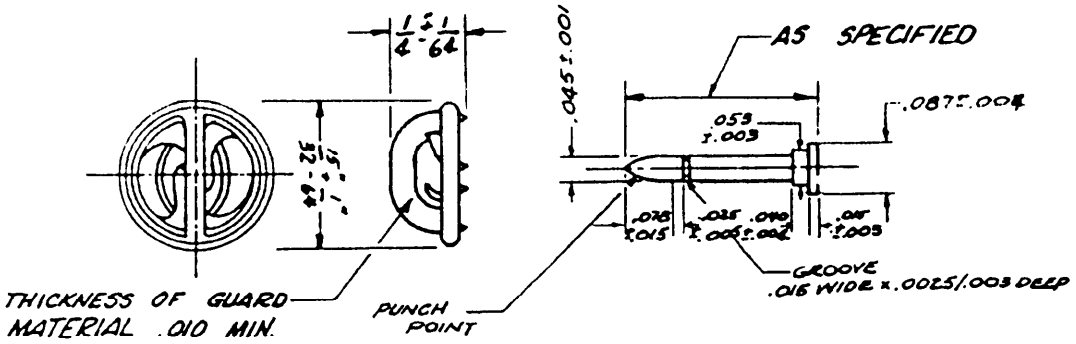
Custodian:
Army - IH

Preparing activity:
Army - IH

Review activity:
DLA - CT

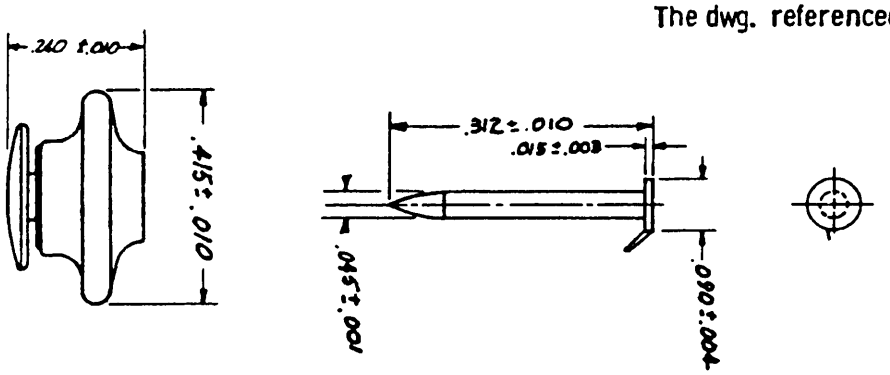
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CLUTCH PRONG

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This copy is furnished for information only.
The dwg. referenced in para. 2.1 governs.



FLAT BALL TYPE CLUTCH PRONG WITH PICK

FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-1563C	2. DOCUMENT DATE (YYMMDD) 901231

3. DOCUMENT TITLE
BUTTONS, INSIGNIA AND FINDINGS (U. S. M. A.), GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
The Institute of Heraldry	(202) 274-6636	284-6636
c. ADDRESS (Include Zip Code) Bldg 15, Technical & Production Division Cameron Station	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466	

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